Problem Chosen	2026 MCM/ICM Summary Sheet	Team Control Number
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Abstract

At present, the WNBA and its teams are undergoing rapid transformation, necessitating innovation in traditional sports management approaches. This study takes the Indiana Fever of the WNBA as an empirical case to construct an integrated dynamic decision-making framework for optimizing a team's long-term value, aiming to address the core challenge of balancing competitive objectives with financial sustainability in professional sports franchises.

We model multi-period operations as a sequential decision-making process using dynamic programming theory. Building upon this foundation, we systematically develop four sub-models: player selection and optimization, league expansion evaluation, ticket pricing optimization, and injury response. These sub-models comprehensively support strategic decisions concerning on-court performance, commercial operations, and financial sustainability.

First, in modeling commercial value, we establish a dynamic programming core model based on the Bellman equation, achieving the optimization of long-term discounted value through the interaction mechanism of state and decision variables. For modeling competitive value, we introduce a logistic response function to characterize the nonlinear relationship between team strength and game win probability. Parameter sensitivity analysis shows that the positive correlation between competitive strength and win probability remains stable within the range $\beta \in [0.05, 0.25]$.

Second, the study constructs a mixed-integer programming model for roster optimization, simultaneously weighing players' competitive contributions and commercial value under salary budget and roster size constraints. Results indicate that core player selection is consistently dominated by competitive value across a commercial weight coefficient range of $\beta \in [0, 2]$. To assess the external environmental change of league expansion, we develop an evaluation model incorporating market overlap and travel cost coefficients. Three-dimensional parameter space simulations reveal that a positive value change occurs only when a single-team expansion is implemented and the new team enters a region with low market overlap ($O_L < 0.5$). The ticket pricing optimization model identifies that the team's optimal price should be approximately 15-20% higher than the price that maximizes short-term revenue, with long-term fan lifetime value contributing up to 35% of total value. Quantitative analysis of injury shocks indicates that if a core player is completely unavailable, the team's commercial revenue loss could be 2.3 times greater than its ticket revenue loss.

The research provides a decision framework for the Indiana Fever: they should capitalize on the current strategic window to implement targeted player acquisition and moderate financial leverage expansion; advocate for a single-team, gradual expansion model with complementary geographic placement during league growth; establish a long-term fan asset-oriented pricing system; and formulate injury response plans focused on protecting commercial value. The robustness of the vast majority of conclusions is verified through systematic sensitivity analysis.

Keywords: Dynamic Decision Model; Professional Sports Management; Value Optimization; Bellman Equation

February 3, 2026

Contents

1 Introduction

1.1 Problem Background

The key to successful sports management lies in balancing financial sustainability with competitive excellence. In the face of fierce market competition and high operational costs, championship honors alone cannot sustain an organization's long-term development. Conversely, pursuing profits at the expense of athletic performance will erode brand value.

Currently, many leagues—represented by the Women's National Basketball Association (WNBA) —are undergoing rapid transformation. Factors such as increased audience engagement, expanded media contracts, and the redefinition of player rights collectively create an environment brimming with both opportunities and risks. However, traditional sports management models reveal significant flaws in this context: players, as core assets, often have their comprehensive value measured by single metrics or short-term gains. Additionally, management frequently relies on massive investments such as high transfer fees and player salaries to secure short-term competitive success. This approach leaves teams vulnerable to market fluctuations and unforeseen risks, as their decision-making often lacks scientific and systematic support. Consequently, developing a decision-making model capable of responding to both athletic performance and commercial operational changes has become crucial for teams seeking long-term success.

1.2 Restatement of the Problem

Based on the background information and problem constraints, the following tasks must be completed:

1. Design a dynamic decision-making model to maximize team profit and long-term value.
2. Based on the team's needs and the model, develop a strategy for acquiring players in the upcoming season.
3. Analyze how the team's strategy should adapt when the league expands.
4. Formulate an optimal ticket pricing strategy for the upcoming season based on the model.
5. Analyze the impact of incorporating player injuries on team strategy.

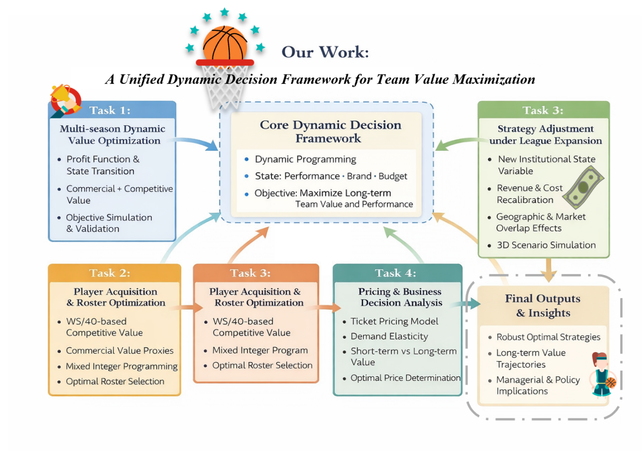
1.3 Our Work

2 Assumptions and Explanations

2.1 Assumptions

Considering that practical problems always contain many complex factors, first of all, we need to make reasonable assumptions to simplify the model:

- Assumption 1 Each player can perform to their full potential without interference, and their actual performance is consistent with their historical efficiency metrics (e.g., WS/48).



- Assumption 2: The team management makes decisions at the beginning of each season. These decisions affect the current season's revenue and costs, as well as the team's state in the next season.
- Assumption 3: It is assumed that during the research period, there will be no structural changes in league rules, the salary cap system, the revenue-sharing structure, or the overall market environment.

2.2 Notations

Some important mathematical notations used in this paper are listed in the following table.

Table 1: The List of Key Notations

Notation	**Description**
$S_t = (W_t, M_t, B_t)$	State vector
$A_t = (\alpha_t, \beta_t, P_t, \lambda_t)$	Decision vector
$V(S_t)$	Bellman value function
γ	Discount factor ($\gamma \in (0, 1)$)
TeamWS_t	Team strength index (WS/40 weighted)
β	Logistic coefficient
O_L	Market overlap ($O_L \in [0, 1]$)
τ	Travel cost coefficient
ΔN	Expansion scale ($\Delta N = +1, +2$)
ΔV	Value change
LTV	Customer lifetime value

2.3 Data Preprocessing

2.3.1 Data Collection

This study constructs an integrated dataset for WNBA expansion and competitiveness modeling by combining publicly available league, team, and player data from multiple verified sources. Most structured datasets were downloaded from Kaggle and Basketball-Reference, with upstream original sources traced explicitly to ensure transparency and reproducibility.

2.3.2 Data Processing

After data collection, all datasets were systematically processed to ensure temporal consistency, completeness, and suitability for modeling.

First, data from the three target teams—Indiana Fever, Atlanta Dream, and Seattle Storm—were merged into a single unified table. Team-level datasets originally stored separately were concatenated using consistent team identifiers, with variables aligned by season to form a panel-style structure suitable for longitudinal analysis.

Second, the temporal scope of the data was strictly restricted to the 2022–2025 seasons. Observations outside this range were removed to maintain consistency across teams and to ensure that all variables used in the analysis correspond to the same competitive and market environment.

Third, incomplete team-level records were identified for the 2023–2024 seasons, where certain financial or attendance-related variables were missing for specific teams. These missing values were handled programmatically using Python. For each affected variable, missing observations were imputed using the season-level mean calculated across available teams. This approach preserves overall distributional characteristics while avoiding distortions caused by listwise deletion.

3 Question 1: Construction of Dynamic Decision-making Model

3.1 Sub-model I: Commercial Value Model

The model employs **dynamic programming** as its core methodology, treating a team's multi-season operations as a sequential decision-making process. At the start of each season, management makes decisions based on the team's current state. These decisions impact the current season's profit and influence future seasons' states through state transition functions. By defining a **value function** and **Bellman equations**, the model transforms the long-term profit maximization problem into a recursive optimization problem, thereby seeking optimal strategies in a stochastic environment.

3.1.1 Bellman Optimality Equation

The goal of team management is to maximize the sum of discounted future profits. Defining the value function $V(S_t)$ as the maximum expected profit starting from season t in state S_t, we have the Bellman equation:

$$V(S_t) = \max_{A_t}\{\Pi_t(S_t, A_t) + \gamma \mathbb{E}[V(S_{t+1}) \mid S_t, A_t]\}$$

where:

- $\Pi_t(S_t, A_t)$ is the current-period profit generated by taking decision A_t in state S_t.
- $\gamma \in (0, 1)$ is the discount factor, balancing short-term profits and long-term gains;
- $\mathbb{E}[\cdot]$ denotes the expectation over uncertainties.

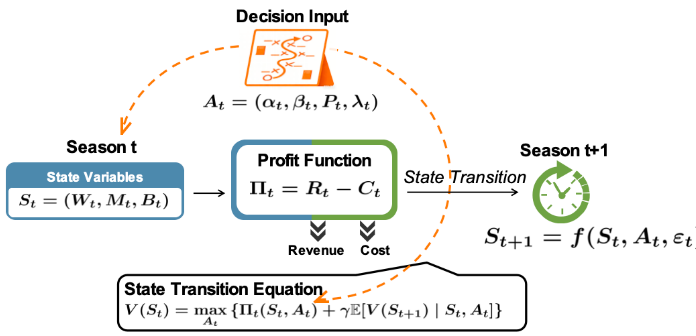


Figure 1: Flow chart of the core mechanism of the dynamic model

3.1.2 State Variables and Decision Variables

The **state vector** $S_t = (W_t, M_t, B_t)$ summarizes the team's core characteristics:

- W_t: Sporting performance level (e.g., win rate), which directly influences revenue.
- M_t: Market and brand value, which determines commercial revenue potential.
- B_t: Financial status or available budget, which constrains cost expenditures and investment capacity.

The **decision vector** $A_t = (\alpha_t, \beta_t, P_t, \lambda_t)$ represents the management's control levers at the beginning of each season:

- α_t : Total player salary expenditure, which determines the quality of player recruitment.
- β_t : Operations and marketing expenditure, which used for digital platforms, merchandise development, and fan engagement.
- P_t : Average ticket pricing strategy, directly intervenes in venue/gate revenue.
- λ_t : Financing leverage ratio, proportion of new expenditures financed by debt.

The decision A_t directly affects the current profit Π_t and influences the future state through the **state transition equation** $S_{t+1} = f(S_t, A_t, \epsilon_t)$, where ϵ_t is a random disturbance.

3.1.3 Profit Function Model

The profit function is the core component linking decisions to current-period financial outcomes, defined as total revenue minus total cost:

$$\Pi_t = R_t - C_t$$

where:

- Π_t: the team's profit in season t;

- R_t: the total revenue in that season;
- C_t: the total cost in that season.

The following is the decomposition diagram of state variables, decision variables and profit function in the model, which explains the specific composition of the model and the meaning of parameters

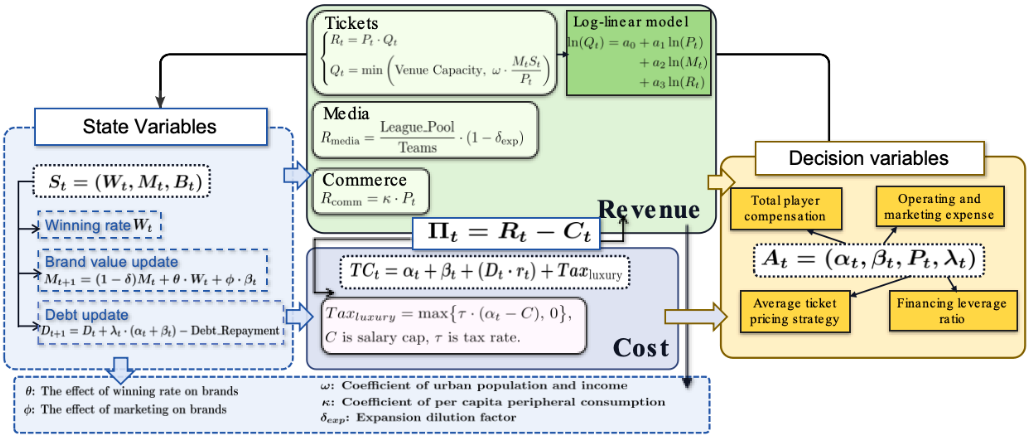
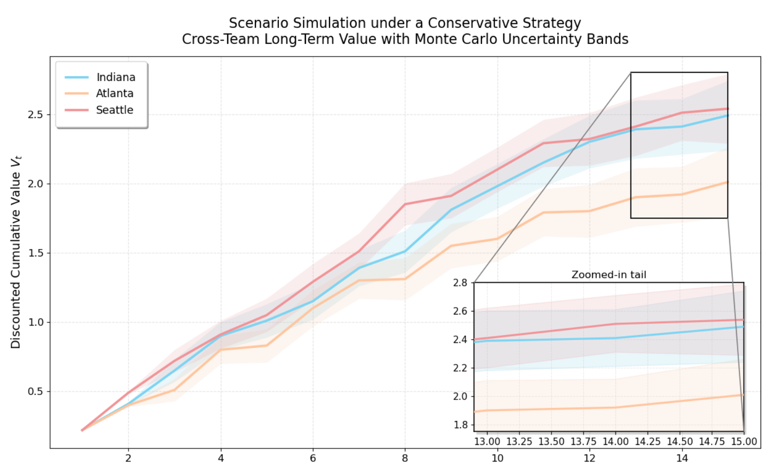


Figure 2: Decomposition diagram of state variables, decision variables and profit function

3.1.4 Scenario Simulation and Model Verification

To validate the model's effectiveness, this study conducted scenario simulations based on a conservative management strategy (risk-controlled, long-term sustainable, and independent of extreme assumptions).



Under uniform strategic conditions, simulation results reveal significant differences in the long-term value evolution of different teams. For instance, teams in large markets with high audience capacity (Indiana Fever, Seattle Storm) typically exhibit more stable long-term growth trends under identical strategies, whereas small-market teams (Atlanta Dream) are more susceptible to random fluctuations. This indicates the model captures structural differences between teams rather than attributing value changes solely to management decisions. Furthermore, the long-term value rankings of teams remained stable across the range of random fluctuations covered by the Monte Carlo simulations, further demonstrating the model's robustness in handling uncertainty. Simultaneously, the simulation results reflected the economic rationality of long-term decision-making: value growth exhibited nonlinear characteristics consistent with the economic principles of diminishing marginal returns on investment and time discounting.

3.2 Sub-model II: Competitive Performance Model

This model aims to quantify each player's marginal contribution to team's winning rate in a particular season, converting this contribution into a comparable "competitive monetary value" $V_{competitive}$.

Firstly, the model employs **"Win Shares per 40 minutes" (WS/40)** to measure individual player's contribution. Player's contribution is then weighted by playing time to derive the team strength metric WS_i. Finally, a **Logistic link function** maps team strength to single-game win probability. This step enables us to quantify each player's marginal contribution to total season's winning, predicting the ability to win for the next season.

3.2.1 Time-weighted WS/40

To quantify individual player performance, we use WS/40, a widely used advanced basketball metric that estimates the number of team wins contributed by a player per 48 minutes of play. Win Shares decomposes team success into offensive and defensive components and allocates these contributions to individual players based on box-score statistics and team efficiency.

The team strength index WS_i aggregates contributions from all players on the roster. This study employs WS/40 as the core player metric. Standardized by playing time, this metric enables fair comparison of marginal contributions to team wins across different player's roles. Its calculation is transparent and directly aligned with the objective of winning.

$$\text{TeamWS} = \sum_{i \in \text{roster}} \text{WS}/40_i \times \frac{\text{Minutes}_i}{40}$$

3.2.2 Logistic Linkage Function

When predicting specific winning rates, the model incorporates a Logistic linkage function. This design ensures winning rate outcomes remain within the reasonable range of 0 to 1 while reflecting the nonlinear characteristics common in competitive sports: when a team's strength is already exceptional, the winning rate improvement from adding top-tier players gradually slows—demonstrating diminishing marginal returns. This approach naturally simulates the "$1 + 1 < 2$" phenomenon in team sports, enabling precise calculation of each player's actual incremental contribution to total season wins. This forms the basis for reasonably pricing a player's competitive value.

$$Y = \begin{cases} 1, & \text{(Win)} \\ 0, & \text{(Loss)} \end{cases}$$

$$\Pr(Y = 1) = \frac{1}{1 + e^{-(\alpha+\beta\cdot\text{TeamWS})}}$$

- α: Basic win rate (league average)
- β: The marginal impact of a team's winning potential on the winning rate
- TeamWS: Team strength built by player efficiency

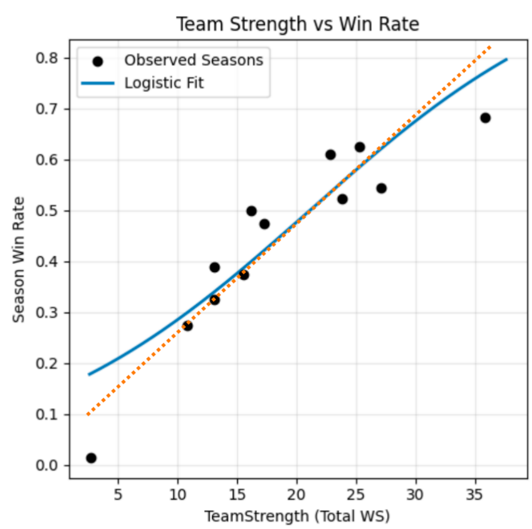
By fitting four seasons of data to three teams, we obtained $\begin{cases} \text{Calibrated } \alpha : -1.7447 \\ \text{Calibrated } \beta : 0.0825 \end{cases}$

After obtaining the single-game win probability p, it can be transformed into the expected win count for the entire regular season. Let the total number of games in the season be G (for the WNBA regular season, typically $G \approx 40$), then the team's expected number of wins in that season is:

$$E[\text{Wins}] = p \times G$$

3.2.3 Model Conclusions and Player Marginal Contribution Analysis

To visually illustrate the nonlinear relationship between team strength and win probability, we plotted the following figure.



The graph clearly shows that as Team Strength increases, the season win rate exhibits a roughly "S-shaped" growth curve. This trend perfectly validates the applicability of the Logistic function and reveals the nonlinear characteristics of players' marginal contributions:

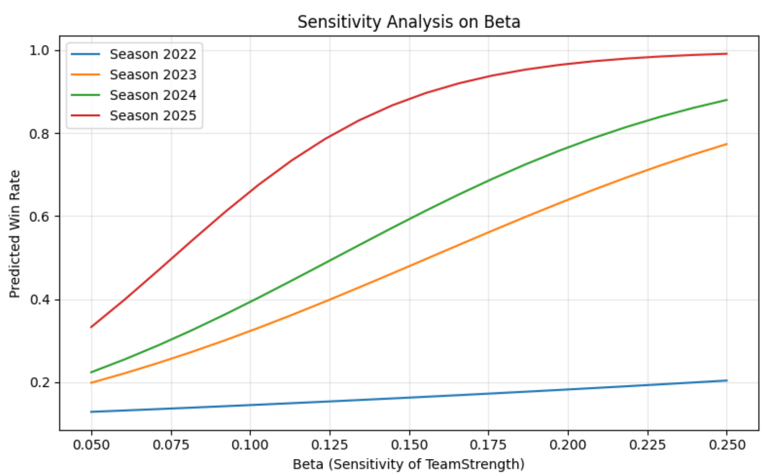
- Low-Strength Range: The curve is flat, indicating weak marginal impact of strength improvements on win rate. This suggests weak teams require structural rebuilding, as adding a single star player is unlikely to significantly improve performance.

- Medium-Strength Range: The curve has its steepest slope, representing the "high-efficiency zone" where strength translates into win rate. Teams in this range achieve the most substantial win rate gains by introducing key players, making it the most cost-effective reinforcement phase.

- High-strength range: The curve approaches its upper limit, showing clear diminishing marginal returns. Top-tier teams experience reduced returns from further stacking star players.

3.2.4 Parameter Sensitivity Analysis

Due to the limited data sample in this study, model parameter estimates may involve uncertainty. To validate model reliability, we conducted systematic sensitivity analysis on β, with results shown in the following figure.



This table illustrates how predicted win rates vary across seasons under different β values (Team Strength's sensitivity coefficient to win rate). β ranges from 0.05 to 0.25, covering conservative to aggressive assumptions about strength-to-win conversion efficiency.

The sensitivity analysis indicates that despite uncertainty in the strength coefficient of win Rate, the positive correlation between Team strength and win Rate remains robust across all parameter settings. Meanwhile, the model demonstrates a diminishing marginal returns characteristic in the high team Strength range, suggesting that increased investment does not necessarily yield proportional returns in win rate.

3.3 Comprehensive Model

The final comprehensive value function $V_{\text{total}}(S_t)$ can be expressed as:

$$V_{\text{total}}(S_t) = \lambda \cdot V_{\text{commercial}}(S_t) + (1 - \lambda) \cdot \varphi \cdot V_{\text{competitive}}(S_t)$$

where:

- $V_{\text{commercial}}(S_t)$: Bellman optimal value function of the commercial value model

- $V_{\text{competitive}}(S_t)$: Value output of the competitive performance model (monetizable)

- $\lambda \in [0, 1]$: Strategic preference weight
- φ: Competitive value conversion coefficient (converts win probability into monetary value)

3.4 Strategic Framework

Analysis based on the dynamic programming model and the Logistic competitive return curve indicates that the Indiana Fever's current mid-level competitiveness position (6th in the league in the 2025 season) holds special strategic value. This stage lies within the maximum slope region of the S-shaped growth curve, meaning that each incremental unit of competitive strength (WS/40) the team gains will yield the highest marginal increase in win probability and corresponding commercial returns. This characteristic defines the current period as the **golden strategic window for the franchise to achieve a value leap, shifting the decision-making focus from stable operations to maximizing marginal returns.

3.4.1 Competitive Strategy

Model analysis shows that during this phase, the marginal contribution of overall roster stability is significantly higher than that of individual star power. Therefore, salary allocation should prioritize securing high-contributing players, concentrating salary growth on key players with high WS/40 and high stability in playing time.

3.4.2 Commercial Strategy

Commercial investments should create a resonance effect with improvements in competitive performance. Focus on the narrative theme of "The Rise of a Young Core". Amplify the team's upward momentum by optimizing social media content, developing players' personal brands, and forming targeted partnerships oriented toward the women's sports market.

3.4.3 Financial Strategy

According to the Bellman equation, the expected increase in the discounted future value ($\gamma \cdot \mathbb{E}[V(S_{t+1})]$) from additional investment at this stage will outweigh the immediate financial costs associated with leverage. Therefore, it is appropriate to moderately increase the proportion of debt financing, provided the capital is strictly allocated to projects that directly enhance marginal competitiveness (e.g., signing targeted players) or long-term brand value.

4 Question 2: Player Roster Optimization Model

To support the club in formulating player acquisition strategies for the upcoming season, this study proposes a decision optimization model based on **Mixed-Integer Programming**. The model aims to select an optimal set of players from the free-agent market under realistic budgetary and league regulations, with the objective of maximizing the team's comprehensive value for the new season.

The model defines a player's value as a weighted linear combination of *competitive value* and *commercial value*, thereby simultaneously reflecting the management's dual focus on on-court performance and commercial returns. The weights can be adjusted according to the specific strategic goals of the franchise.

4.1 Decision Model Based On Integer Programming

4.1.1 Decision Variables

The core decision of the model is whether to sign a specific player. For each candidate player i, we define a binary decision variable:

$$x_i = \begin{cases} 1, & \text{if player } i \text{ is signed} \\ 0, & \text{otherwise} \end{cases}$$

All decision variables form a vector $\mathbf{x}$, and a particular combination represents a possible team roster configuration.

4.1.2 Objective Function: Maximizing Total Team Value

The objective is to maximize the total value contributed by the selected players, which is a weighted sum of competitive and commercial contributions:

$$\text{Maximize } Z = \sum_i x_i \left(\alpha \cdot V_i^{\text{competitive}} + \beta \cdot V_i^{\text{commercial}} \right)$$

where:

- $V_i^{\text{competitive}}$ represents the **competitive value** of player i.
- $V_i^{\text{commercial}}$ represents the **commercial value** of player i (quantifiable through metrics such as jersey sales, social media influence, and market appeal).
- Parameters α and β are **preference weights** ($\alpha, \beta \geq 0$) set by the management to balance the relative emphasis on "winning" versus "profitability." For instance, setting $\alpha > \beta$ indicates a stronger preference for building a competitive team, while $\beta > \alpha$ prioritizes commercial appeal.

4.1.3 Core Constraints

The optimization must satisfy the following realistic constraints:

1. **Budget Constraint**:
$$\sum_i c_i x_i \leq B$$
where c_i is the estimated acquisition cost (e.g., annual salary) of player i, and B is the total available team budget. This ensures the total salary does not exceed the salary cap or the owner's financial limit.

2. **Roster Size Constraint**:

$$N_{\text{min}} \leq \sum_i x_i \leq N_{\text{max}}$$

This ensures the final roster size complies with the league's minimum (N_{min}) and maximum (N_{max}) player requirements.

4.1.4 Measuring Competitive Value

The competitive value $V_i^{\text{competitive}}$ in the model is measured using WS/40.
It is calculated as follows:

$$V_i^{\text{competitive}} = WS_i = \frac{(WS/40)_i \times \text{Minutes}_i}{40}$$

4.1.5 Measuring Commercial Value

In professional sports, a player's commercial value manifests in their ability to attract fan attention, boost media viewership, and drive merchandise sales. To incorporate this into the optimization model, this study adopts a linear additive model quantified by three standardized proxy variables:

$$V_i^{\text{commercial}} = w_1 \cdot \text{SocialMedia}_i + w_2 \cdot \text{AllStar}_i + w_3 \cdot \text{DraftValue}_i$$

- **Social Media Index**: To prevent the extreme popularity of a few superstars from dominating the model, the logarithm of the follower count is taken to capture the diminishing marginal effect of personal brand influence:

$$\text{SocialMedia}_i = \log(\text{Followers}_i + 1)$$

- **All-Star Indicator**: A binary variable serves as a proxy for public popularity and market appeal:

$$\text{AllStar}_i = \begin{cases} 1, & \text{if player } i \text{ has been an All-Star in recent years} \\ 0, & \text{otherwise} \end{cases}$$

- **Draft Premium Decay Term**: To reflect how the commercial premium associated with high draft picks diminishes over a career, being gradually replaced by actual performance, the following decay model is constructed:

$$\text{DraftValue}_i = \frac{1}{\text{DraftPick}_i} \cdot \exp(-\delta \cdot \text{YearsInLeague}_i)$$

  where $\delta > 0$ is the decay coefficient, controlling the speed of premium attenuation.

- **Parameter Specification and Sensitivity Analysis**

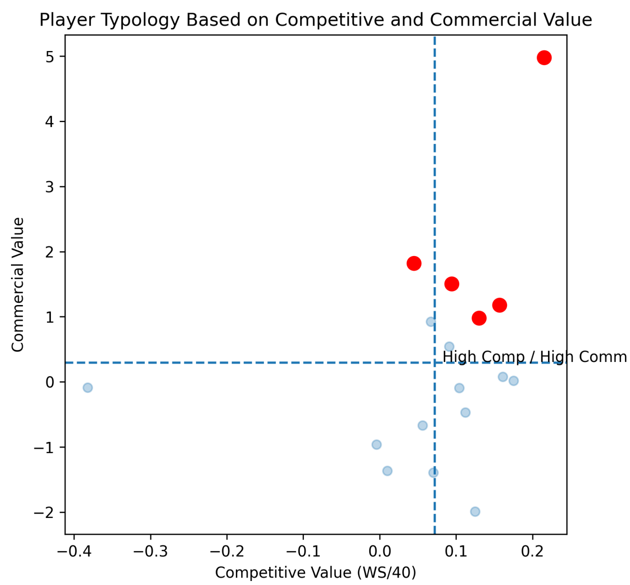
  It is suggested to set baseline weights as $w_1 = w_2 = w_3 = 1$. The decay coefficient δ and all weight parameters will be included in a systematic **sensitivity analysis**. This analysis aims to explore the impact of different commercial strategies (e.g., prioritizing immediate market influence versus long-term potential investment) on the final roster selection.

4.2 Strategy Analysis

Based on the integer programming model, we aim to maximize comprehensive value and select the optimal combination of five players from the Indiana Fever roster. We further conduct a two-dimensional profile analysis of the selected players' athletic and commercial characteristics.

By dividing candidate players into quadrants based on sample means, they can be classified into four categories:

- **High Athletic – High Commercial**
- **High Athletic – Low Commercial**
- **Low Athletic – High Commercial**
- **Low Athletic – Low Commercial**

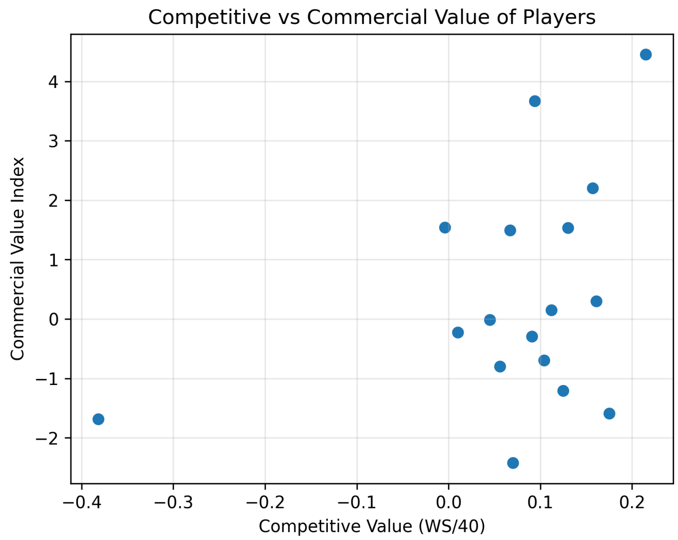


The results indicate that the selected players are not extreme optima in a single dimension; instead, they achieve a more balanced configuration between athletic performance and commercial value.

The model outcomes provide decision support for player signings in the new season: the optimal signing combination prioritizes Category 1 players and, when necessary, supplements with a small number of Category 2 players to enhance athletic depth. Category 3 influencer-type players are approached with caution. These results reflect a rational decision-making preference, where teams are more inclined to allocate limited resources to assets with the highest long-term comprehensive returns.

4.3 Dimensional Independence Analysis

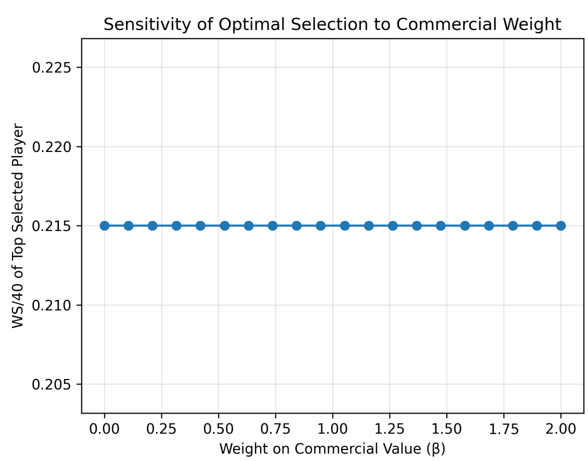
To examine the structural rationality of the model, we analyzed the correlation between athletic value and commercial value. The following scatter plot illustrates the distributional relationship between the two.



The data reveal no significant linear correlation (with a correlation coefficient close to 0) between a player's WS/40 metric and the standardized commercial value index, presenting a typical weak-correlation distribution. This finding indicates that athletic performance and commercial influence represent distinct dimensions of a player's value, each providing non-substitutable informational value.

4.4 Parameter Sensitivity Analysis: Robustness of Core Decision-Making

Having validated the model's structural rationality, we further investigated the impact of the commercial value weight coefficient (β) on the model's decision-making. The figure below shows the variation in the model's optimal player selection as β changes within the interval $[0, 2]$.



The results indicate that regardless of how is adjusted, the athletic performance metrics of the players selected as optimal consistently remain at a specific high level. This shows that within the tested parameter range, commercial variables have almost no effect on the selection of top-tier players but may influence the ranking of mid-level and marginal players. The model's core decision-making is entirely driven by athletic performance. This decision-making pattern reflects a clear strategic logic: teams should establish an “athletic performance first” selection framework, using commercial value as a “tie-breaker” among players with comparable athletic ability. Specifically, the construction of the core rotation lineup should be led by athletic value to ensure the team’s competitive baseline, whereas the selection of bench players and market-oriented signings can incorporate greater consideration of comprehensive value, achieving a balance between athletic and commercial objectives.

5 Question3: Dynamic Strategy Model and Location Analysis Under League Expansion

Professional sports league expansion alters the competitive ecosystem, revenue distribution, and operating costs for existing franchises, impacting their long-term value and strategic viability. Building on the dynamic programming model from Question 1, we introduce institutional variables with minimal necessary extensions to construct a quantitative decision-analytic framework for assessing the impact of expansion.

5.1 New State Variable: Institutional Status

To characterize the institutional event of expansion, we add a binary variable to the original state vector:

$$r_t \in \{0, 1\}$$

where $r_t = 0$ represents the baseline non-expansion regime, and $r_t = 1$ represents the post-expansion regime. This variable acts as a switch, controlling the adjustments to the revenue and cost parameters described below.

5.2 Expansion-Adjusted Model Specifications

5.2.1 Expansion-Adjusted Revenue Function

Each component of the total revenue function $TR_t(r_t)$ is affected by the institutional status r_t and the geographic characteristics of the new franchise(s). Specifically:

- **Ticket Revenue**: The market response coefficient ω in the attendance demand function is modified as:

  $$\omega(r_t) = \omega_0 \cdot (1 - \eta \cdot O_L \cdot r_t)$$

  where $O_L \in [0, 1]$ is the **market overlap** between the new team and the target team, and η is the demand diversion intensity coefficient. This equation quantifies the erosion effect of proximate competition on the local fan base.

- **Media Revenue Share**: Both the league’s total media revenue pool LP and the number of teams N change with expansion:

  $$\text{Rev}_t^{\text{media}}(r_t) = \frac{LP(r_t)}{N(r_t)} \cdot (1 - \delta_{\text{exp}}), \quad \begin{cases} N(r_t) = N_0 + \Delta N \cdot r_t \\ LP(r_t) = LP_0 \cdot (1 + g_{LP} \cdot r_t) \end{cases}$$

Here, ΔN is the number of new teams, and g_{LP} is the growth rate of the league's total media revenue attributed to expansion.

- **Commercial Revenue**: Considering the impact of star appeal on sponsorship and merchandise sales, its growth coefficient is adjusted:

$$\kappa(r_t) = \kappa_0(1 - \eta_c O_L r_t)$$

Typically, commercial revenue is more sensitive to star effects, hence $\eta_c \geq \eta$.

5.2.2 Expansion-Adjusted Cost Function

The total cost function $TC_t(r_t)$ is affected by expansion in the following aspects:

- **Player Salary Inflation**: Expansion intensifies competition for scarce top-tier talent, leading to an increase in league-wide salary levels. The player salary cost component is adjusted as:

$$\alpha_t(1 + \kappa_{FA} \cdot r_t)$$

where κ_{FA} is the **free agency premium coefficient**.

- **Schedule and Travel Costs**: The new geographic location L may increase away-game travel distance, raising associated costs:

$$C_{\text{travel}} \cdot \tau \cdot r_t$$

where τ is the **travel increment coefficient** associated with location L.

5.3 State Transition and Value Function

In the state transition equation for brand value M_t, the current win rate W_t is considered a function of r_t, denoted as $W_t(r_t)$, to reflect changes in competitive balance post-expansion. Integrating the above adjustments, the Bellman optimality equation under the expansion regime is:

$$V(S_t, r_t) = \max_{A_t} \{[TR_t(r_t, O_L) - TC_t(r_t, \tau)] + \gamma \, \mathbb{E}\,[V(S_{t+1}, r_{t+1}) \mid S_t, A_t, r_t]\}$$

The transition of the institutional state from $r_t = 0$ to $r_t = 1$ is modeled as a deterministic event occurring in a specific season.

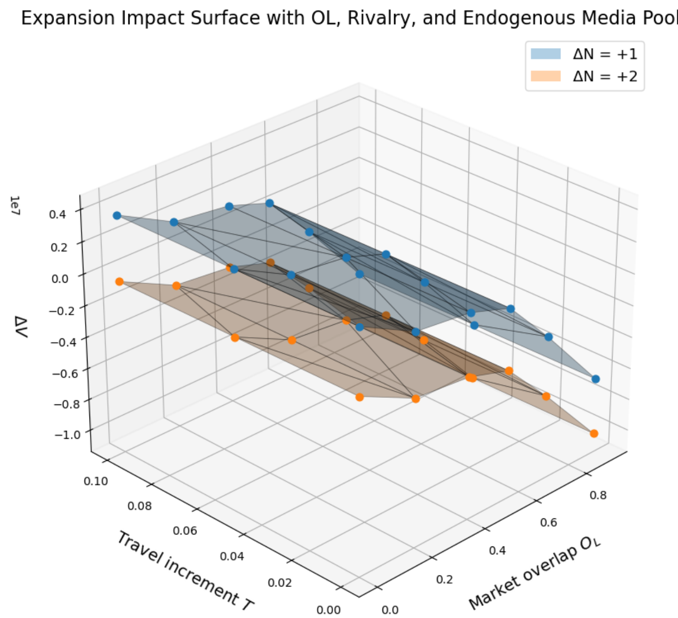
5.4 Scenario Simulation: Three-Dimensional Parameter Space Analysis

To comprehensively evaluate the heterogeneity of expansion impacts, we designed a systematic simulation based on a three-dimensional parameter space.

- **Expansion Scale** (ΔN): Set to two scenarios: +1 and +2, representing the number of new teams.
- **New Team Geographic Characteristics**:
  - **Market Overlap** (O_L): Uniformly sampled in the interval [0, 1] with a step of 0.1.

– **Travel Increment Coefficient** (τ): Uniformly sampled in [0, 1], quantifying the additional costs due to geographic distance.

For each parameter combination $(\Delta N, O_L, \tau)$, we solve for the optimal post-expansion strategy π_L^* and its corresponding long-term value V_L. This is compared with the baseline value V_0 to calculate the value change $\Delta V(O_L, \tau) = V_L - V_0$.



5.5 Results Analysis and Managerial Implications

Simulation results reveal that:

- Limited single-team expansion ($\Delta N = +1$) can create net value for the team ($\Delta V > 0$), whereas overly rapid two-team expansion ($\Delta N = +2$) leads to value destruction ($\Delta V \leq 0$) due to talent dilution, revenue thinning, and operational complexity.
- The surface shows a steep decline in regions where $O_L > 0.6$, confirming that significant market overlap (e.g., within the same city or core fan market) is highly detrimental.
- The surface exhibits an approximately linear decline along the τ-axis, indicating that increased travel costs exert a continuous and predictable erosion on franchise value.

We can infer that locating a new franchise in a city with low-to-moderate market overlap ($O_L \approx$ 0.2–0.5) and geographic proximity (low τ) would maximize franchise profit. Conversely, locating it in a market with high overlap with the target team ($O_L > 0.7$, such as the same city) or a geographically remote area (high τ) would induce severe diversion effects and cost burdens.

6 Question 4: Pricing Model

Ticket prices not only affect single-game ticket sales revenue but also profoundly influence a team's long-term financial health and brand value through their impact on attendance, fan experience, and long-term fan conversion. This section aims to establish a quantitative analytical framework to explore the optimal pricing strategy that maximizes the total seasonal value.

6.1 Composition of Pricing Model

Building upon the profit structure from Question 1, we construct a ticket pricing model that decomposes the total seasonal value into immediate and long-term components.

Total Seasonal Value $V(P)$ = Immediate Ticket Revenue + Long-Term Season Ticket Value

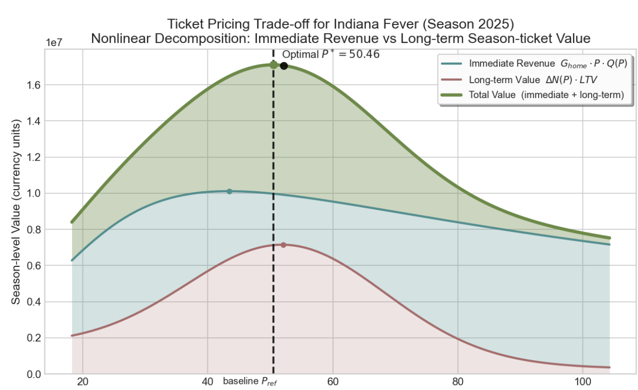
$$V(P) = G_{\text{home}} \cdot P \cdot Q(P) + \Delta N(P) \cdot \text{LTV}.$$

- G_{home}: Number of home games (constant).
- $Q(P)$: **Attendance Rate Function**.
- $\Delta N(P)$: **Number of Newly Converted Season Ticket Holders**. Calculated by:

$$\Delta N(P) = \eta(P) \cdot G_{\text{home}} \cdot Q(P).$$

- $\eta(P)$: **Price-Dependent Conversion Rate Function**. Its peak is set within the medium price range, reflecting higher loyalty and conversion potential for audiences in that price bracket.
- LTV: **Lifetime Value of a Season Ticket Holder**. Calculated using a discounted cash flow model, representing the present value of long-term revenue.

6.2 Analysis of Results: Optimal Price Determination



The immediate revenue curve exhibits a typical single-peak structure, reflecting the trade-off between rising prices and declining attendance. The long-term revenue curve also shows nonlinear characteristics, peaking around medium price levels, which corresponds to the zone of highest fan conversion efficiency. The total value curve, as the sum of the two, displays a clear interior optimal solution. The optimal ticket price P^* is precisely achieved at the balance point of these two effects and aligns closely with the real-world benchmark price.

7 Question 5: Dynamic Decision Framework Under Injury Shocks

Building upon the original comprehensive value model, this study introduces an injury shock variable to construct a dynamic decision framework capable of quantifying the impact of injuries on a team's competitive and commercial value. By adjusting team strength assessment and revenue functions, this framework enables management to make optimal resource allocation decisions under injury risk.

7.1 Injury Shock Variable

To systematically characterize injury impact, we define a continuous injury shock variable:

$$I_{k,t} \in [0, 1]$$

where:

- k: Identifier for a core player
- t: Season or decision period
- $I_{k,t} = 0$: Player is fully healthy
- $I_{k,t} = 1$: Player is completely unavailable
- Intermediate values indicate partial availability or injury severity

This variable transforms discrete injury states into a continuous measure, providing a finer-grained reflection of injury impact on player availability.

7.2 Injury-Adjusted Competitive Value Model

Team competitive strength is measured based on players' Win Shares per 40 minutes (WS/40). When a core player is injured, their marginal contribution to team strength is reduced proportionally to their absence:

$$\text{TeamStrength}_t = \sum_{i \neq k} \text{WS}_{i,t} + (1 - I_{k,t}) \cdot \text{WS}_{k,t}$$

The adjusted team strength is then mapped to expected wins via a logistic function:

$$\text{TeamWS}_t = \sum_{i \neq k} \left[\text{WS/40}_{i,t} \times \frac{\text{Minutes}_{i,t}}{40} \right] + (1 - I_{k,t}) \cdot \left[\text{WS/40}_{k,t} \times \frac{\text{Minutes}_{k,t}}{40} \right]$$

where G is the total number of games in the season. This adjustment captures the non-linear decay of competitive capability due to injuries.

7.3 Injury-Adjusted Commercial Value Model

The total revenue function is decomposed into three components with varying sensitivities to injury:

7.3.1 Ticket Revenue Adjustment

Attendance is influenced by both team performance and star appeal:

$$Q_t = \min\left(\text{Cap}_{\text{venue}}, \omega \cdot \frac{M_t}{P_t}\left[\alpha R_t + (1-\alpha)(1-\lambda_{\text{att}} I_{k,t})\right]\right)$$

where:

- R_t: Expected number of wins (competitive appeal)
- λ_{att}: Sensitivity coefficient of core player absence on attendance
- α: Relative weight of competitive performance versus star effect

7.3.2 Media Revenue Adjustment

League broadcast revenue is determined by long-term contracts; in the short term, core player injuries have limited impact on an individual team's media share:

$$\text{Rev}_{\text{media},t} = \frac{\text{League Pool}}{\text{Teams}}(1-\delta_{\text{exp}})$$

This assumption reflects the buffering effect of league revenue-sharing mechanisms against injury shocks.

7.3.3 Commercial Revenue Adjustment

Commercial revenue (sponsorship, merchandise, etc.) is more sensitive to individual star brand value:

$$\text{Rev}_{\text{comm},t} = \kappa_1 R_t + \kappa_2(1-\lambda_{\text{comm}} I_{k,t})$$

where $\lambda_{\text{comm}} > \lambda_{\text{att}}$, indicating that commercial revenue is more sensitive to star absence than ticket revenue.

7.4 Extension of the Comprehensive Value Model

The injury shock is integrated into the original comprehensive value model:

$$V_{\text{total}}(S_t, I_{k,t}) = \lambda \cdot V_{\text{commercial}}(S_t, I_{k,t}) + (1-\lambda)\cdot\varphi\cdot V_{\text{competitive}}(S_t, I_{k,t})$$

where:

- $V_{\text{commercial}}$: Solution to the Bellman equation considering injury-adjusted revenues
- $V_{\text{competitive}}$: Competitive value based on adjusted team strength
- The state transition function $S_{t+1} = f(S_t, A_t, I_{k,t}, \epsilon_t)$ now includes the injury shock

7.5 Strategy Analysis

Based on the scenario simulation results, we propose the following adaptive management strategy recommendations:

- **Prioritize Stabilizing Brand and Commercial Partnerships Over Short-Term Ticket Sales.** Given the higher sensitivity of commercial revenue to injuries, management should prioritize maintaining sponsor relationships and brand exposure after a core player is injured.
- **Distinguish Between Athletic Recovery and Commercial Hedging Strategies.** While tactical adjustments can partially restore win totals, they are unlikely to fully compensate for commercial losses. Therefore, independent market and communication strategies are required.

8 Strength and Weakness Analysis

8.1 Strength

Our model frames team management as a dynamic, forward-looking decision problem, explicitly linking current actions with future outcomes. By incorporating state transitions and long-term value maximization, it avoids short-sighted strategies and better reflects real-world sports management practices. In addition, the framework jointly considers competitive performance and commercial value through a weighted objective function, preserving interpretability while allowing flexible strategic emphasis. Scenario analyses further show that the model's core strategic recommendations remain stable under various uncertainties-such as league expansion, pricing effects, and player injuries-demonstrating its robustness as a decision-support tool.

8.2 Weaknesses

Despite these strengths, several limitations remain. Limited data availability reduces the statistical robustness of some estimated parameters, particularly in the win probability model. Moreover, player evaluation focuses on individual metrics and does not fully capture positional balance or team complementarities. Injury effects are treated as short-term exogenous shocks, overlooking long-term risk accumulation, recovery dynamics, and preventive factors.

8.3 Conclusion

Overall, these limitations point to several useful extensions. Parameter uncertainty could be integrated via stochastic simulations or sensitivity analysis, allowing assessment across a range of scenarios. Expanding the dataset with multi-season or player-level information would improve estimation. The win probability model could also be enhanced by including interaction terms or network effects to capture lineup chemistry. Finally, modeling injuries dynamically—linking workload, fatigue, and recovery—would better reflect long-term risk and medical decision-making. These steps would strengthen the model's accuracy and practical relevance.

To:
Herb Simon, Owner, Indian Fever
Amber Cox, General Manager, Indian Fever
From: Team#2628311
Date: February 3, 2026

Dear Mr. Simon and Ms. Cox,

We are pleased to present our recommended strategic framework for the Indiana Fever, designed to support both competitive success and financial sustainability. Our approach is built on a dynamic model that applies the Bellman Optimality Equation to optimize payroll, pricing, marketing, and financial leverage decisions, translating roster strength into win probability and revenue through a WS/40-based logistic function.

Based on the Fever's 6th-place league ranking in the 2025 season, the Fever operates within a medium-competitiveness regime. In this phase, marginal improvements in team strength yield the greatest gains in wins and revenue, offering an ideal window for strategic investment with controlled financial risk.

In this regime, we recommend a precision-focused strategy: payroll growth should target players with strong WS/40 contributions, availability, and core fit, rather than pursuing costly star contracts. Pricing adjustments should be moderated and segmented as performance improves, while marketing efforts ought to highlight upward momentum to build lasting brand value. Financial leverage should only support high-return investments, as excessive debt reduces long-term value in our model.

What's more, to manage uncertainty of the injury of key players, we introduce an injury shock variable: $I_{k,t} \in [0,1]$, where $I_{k,t} = 1$ indicates full health of core player k in season t. This allows the strategy to adapt dynamically to absences. During injuries, priorities shift from aggressive winning to stability—emphasizing roster depth, brand continuity, and sponsor confidence over short-term revenue.

While targeted investment accelerates growth, overcommitment increases risk if performance stalls. Our model therefore includes clear exit conditions: if marginal returns decline or uncertainty rises, spending stabilizes and deleveraging begins to preserve financial flexibility.

In summary, this framework avoids both stagnation and overextension. By leveraging the Fever's current competitive window with disciplined investment, commercial focus, and injury-responsive adjustments, the organization can achieve sustained success while maintaining financial health.

Sincerely,
Team#2628311

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11 Report on Use of AI Tools

1. DeepL (Jan 16, 2026 version) Uploaded part of paper written in Mandarin to be translated into English.

2. OpenAI ChatGPT (Dec 11, 2025 version, ChatGPT-5.2) Query: what is Bellman optimality structure? **Output:** The Bellman optimality structure states that in a multi-stage decision problem, if a policy is optimal overall, then the remaining decisions from any intermediate state must also be optimal for the subproblem starting from that state. Otherwise, replacing the remaining part with a better strategy would improve the overall outcome, contradicting optimality. This property, known as optimal substructure, allows the value of a problem to be defined recursively: the optimal value at the current state equals the maximum immediate reward plus the optimal value of the next state. This recursive relationship is the foundation of dynamic programming and the Bellman equation.

3. OpenAI ChatGPT (Dec 11, 2025 version, ChatGPT-5.2) Query: what's the function of logistic regression **Output:** Logistic regression is used to model and predict the probability of a binary outcome (such as yes/no, success/failure, purchase/not purchase) based on one or more explanatory variables. Its key function is to map a linear combination of inputs into a value between 0 and 1 using the logistic (sigmoid) function, so the output can be interpreted as a probability. This makes logistic regression especially useful for classification, decision boundaries, and interpretable inference about how factors affect the likelihood of an event.

4. OpenAI ChatGPT (Dec 11, 2025 version, ChatGPT-5.2) Query: what's the function of logistic regression **Output:** A player's commercial value is influenced by a combination of on-court performance, personal brand, and market context. From a performance perspective, key factors include playing time, statistical output, awards, team success, and consistency, as these determine visibility and credibility. Star players on winning teams generally attract more sponsorships and media exposure. Beyond performance, personal branding factors play a crucial role. These include social media following and engagement, media presence, personality and public image, marketability (e.g., charisma, storytelling appeal), and off-court behavior. Demographic and contextual elements such as age, nationality, league popularity, team market size, and alignment with sponsor target audiences also significantly affect commercial value.

5. OpenAI ChatGPT (Dec 11, 2025 version, ChatGPT-5.2) Query: what's the function of logistic regression **Output:** Sensitivity analysis is used to evaluate how robust a model's results are to changes in key parameters or assumptions. The basic idea is to vary one or more important inputs—such as costs, demand parameters, discount rates, or weights—while holding others constant, and then examine how the model outputs (optimal decisions, objective values, or policy choices) respond. If the results remain stable over a reasonable range, the model is considered robust. In practice, sensitivity analysis can be conducted through one-at-a-time analysis, multi-parameter variation, or scenario analysis (e.g., optimistic, baseline, and pessimistic cases). For optimization or dynamic programming models, analysts often focus on whether the optimal solution changes discretely, which constraints become binding, and whether the structure of the optimal policy shifts. This process helps demonstrate the reliability and managerial relevance of the model under uncertainty.

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